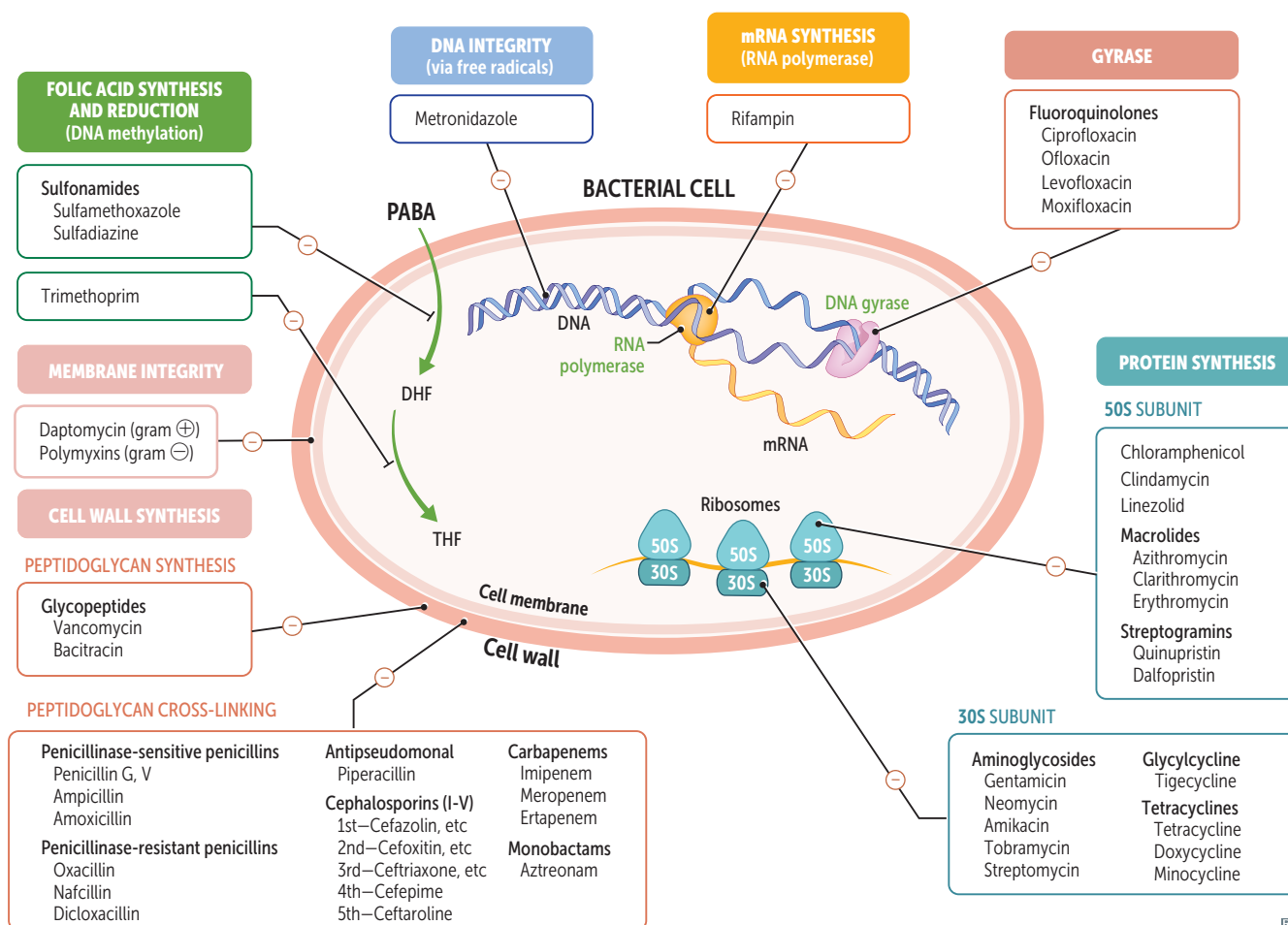


► MICROBIOLOGY—ANTIMICROBIALS

Antimicrobial therapy

**Penicillin G, V**

Penicillin G (IV and IM form), penicillin V (oral). Prototype β -lactam antibiotics.

MECHANISM

D-Ala-D-Ala structural analog. Bind penicillin-binding proteins (transpeptidases). Block transpeptidase cross-linking of peptidoglycan in cell wall. Activate autolytic enzymes.

CLINICAL USE

Mostly used for gram $\oplus$ organisms (*S pneumoniae*, *S pyogenes*, *Actinomyces*). Also used for gram $\ominus$ cocci (mainly *N meningitidis*) and spirochetes (mainly *T pallidum*). Bactericidal for gram $\oplus$ cocci, gram $\oplus$ rods, gram $\ominus$ cocci, and spirochetes. β -lactamase sensitive.

ADVERSE EFFECTS

Hypersensitivity reactions, direct Coombs $\oplus$ hemolytic anemia, drug-induced interstitial nephritis.

RESISTANCE

β -lactamase cleaves the β -lactam ring. Mutations in PBPs.

Penicillinase-sensitive penicillins

Amoxicillin, ampicillin; aminopenicillins.

MECHANISM	Same as penicillin. Wider spectrum; penicillinase sensitive. Also combine with clavulanic acid to protect against destruction by β -lactamase.	Aminopenicillins are amped -up penicillin. Amoxicillin has greater oral bioavailability than ampicillin.
CLINICAL USE	Extended-spectrum penicillin— H influenzae, H pylori, E coli, E nterococci, L isteria monocytogenes, P roteus mirabilis, S almonella, S higella.	Coverage: ampicillin/amoxicillin HHEELPSS kill enterococci.
ADVERSE EFFECTS	Hypersensitivity reactions, rash, pseudomembranous colitis.	
MECHANISM OF RESISTANCE	Penicillinase (a type of β -lactamase) cleaves β -lactam ring.	

Penicillinase-resistant penicillins

Dicloxacillin, nafcillin, oxacillin.

MECHANISM	Same as penicillin. Narrow spectrum; penicillinase resistant because bulky R group blocks access of β -lactamase to β -lactam ring.	
CLINICAL USE	<i>S aureus</i> (except MRSA).	“Use naf (nafcillin) for staph .”
ADVERSE EFFECTS	Hypersensitivity reactions, interstitial nephritis.	
MECHANISM OF RESISTANCE	MRSA has altered penicillin-binding protein target site.	

Piperacillin

Antipseudomonal penicillin.

MECHANISM	Same as penicillin. Extended spectrum. Penicillinase sensitive; use with β -lactamase inhibitors.
CLINICAL USE	<i>Pseudomonas</i> spp., gram $\ominus$ rods, anaerobes.
ADVERSE EFFECTS	Hypersensitivity reactions.

Cephalosporins

MECHANISM	β -lactam drugs that inhibit cell wall synthesis but are less susceptible to penicillinases. Bactericidal.	Organisms typically not covered by 1st–4th generation cephalosporins are LAME : L isteria, A typicals (<i>Chlamydia</i> , <i>Mycoplasma</i>), M RSA, and E nterococci.
CLINICAL USE	1st generation (cefazolin, cephalexin)—gram $\oplus$ cocci, Proteus mirabilis, E coli, Klebsiella pneumoniae. Cefazolin used prior to surgery to prevent <i>S aureus</i> wound infections. 2nd generation (cefaclor, cefoxitin, cefuroxime, cefotetan)—gram $\oplus$ cocci, H influenzae, Enterobacter aerogenes, Neisseria spp., Serratia marcescens, Proteus mirabilis, E coli, Klebsiella pneumoniae. 3rd generation (ceftriaxone, cefpodoxime, ceftazidime, cefixime)—serious gram $\ominus$ infections resistant to other β-lactams. 4th generation (cefepime)—gram $\ominus$ organisms, with $\uparrow$ activity against <i>Pseudomonas</i> and gram $\oplus$ organisms. 5th generation (ceftaroline)—broad gram $\oplus$ and gram $\ominus$ organism coverage; unlike 1st–4th generation cephalosporins, ceftaroline covers MRSA, and <i>Enterococcus faecalis</i>—does not cover <i>Pseudomonas</i>.	1st generation—$\oplus$ PEcK. 2nd graders wear fake fox fur to tea parties. 2nd generation—$\oplus$ HENS PEcK. Can cross blood-brain barrier. Ceftriaxone—meningitis, gonorrhea, disseminated Lyme disease. Ceftazidime for pseudomonaz.
ADVERSE EFFECTS	Hypersensitivity reactions, autoimmune hemolytic anemia, disulfiram-like reaction, vitamin K deficiency. Low rate of cross-reactivity even in penicillin-allergic patients. $\uparrow$ nephrotoxicity of aminoglycosides.	
MECHANISM OF RESISTANCE	Inactivated by cephalosporinases (a type of β -lactamase). Structural change in penicillin-binding proteins (transpeptidases).	
β-lactamase inhibitors	Include C lavulanic acid, A vibactam, S ulbactam, T azobactam. Often added to penicillin antibiotics to protect the antibiotic from destruction by β -lactamase.	CAST (eg, amoxicillin-clavulanate, ceftazidime-avibactam, ampicillin-sulbactam, piperacillin-tazobactam).

Carbapenems

Imipenem, meropenem, ertapenem.

MECHANISM

Imipenem is a broad-spectrum, β -lactamase-resistant carbapenem. Binds penicillin-binding proteins $\rightarrow$ inhibition of cell wall synthesis $\rightarrow$ cell death. Always administered with cilastatin (inhibitor of renal dehydropeptidase I) to $\downarrow$ inactivation of drug in renal tubules.

With imipenem, “the kill is **lastin**’ with **cilastatin**.”

Unlike other carbapenems, ertapenem is not active against *Pseudomonas*.

CLINICAL USE

Gram $\oplus$ cocci, gram $\ominus$ rods, and anaerobes. Wide spectrum and significant adverse effects limit use to life-threatening infections or after other drugs have failed. Meropenem has a $\downarrow$ risk of seizures and is stable to dehydropeptidase I.

ADVERSE EFFECTS

GI distress, rash, and CNS toxicity (seizures) at high plasma levels.

MECHANISM OF RESISTANCE

Inactivated by carbapenemases produced by, eg, *K pneumoniae*, *E coli*, *Klebsiella aerogenes*.

Aztreonam**MECHANISM**

Less susceptible to β -lactamases. Prevents peptidoglycan cross-linking by binding to penicillin-binding protein 3. Synergistic with aminoglycosides. No cross-allergenicity with penicillins.

CLINICAL USE

Gram $\ominus$ rods only—no activity against gram $\oplus$ rods or anaerobes. For penicillin-allergic patients and those with renal insufficiency who cannot tolerate aminoglycosides.

ADVERSE EFFECTS

Usually nontoxic; occasional GI upset.

Vancomycin**MECHANISM**

Inhibits cell wall peptidoglycan formation by binding D-Ala-D-Ala portion of cell wall precursors. Bactericidal against most bacteria (bacteriostatic against *C difficile*). Not susceptible to β -lactamases.

CLINICAL USE

Gram $\oplus$ bugs only—for serious, multidrug-resistant organisms, including MRSA, *S epidermidis*, sensitive *Enterococcus* species, and *C difficile* (oral route).

ADVERSE EFFECTS

Well tolerated in general but **not** trouble **free**: **n**ephrotoxicity, **o**totoxicity, **t**hrombophlebitis, diffuse **f**lushing (**vancomycin infusion reaction** **A**—idiopathic reaction largely preventable by pretreatment with antihistamines and slower infusion rate), DRESS syndrome.

**MECHANISM OF RESISTANCE**

Occurs in bacteria (eg, *Enterococcus*) via amino acid modification of D-Ala-D-Ala to **D-Ala-D-Lac**. “If you **Lack** a **D-Ala** (dollar), you can’t ride the **van** (**vancomycin**).”

Protein synthesis inhibitors

Specifically target smaller bacterial ribosome (70S, made of 30S and 50S subunits), leaving human ribosome (80S) unaffected.

All are bacteriostatic, except aminoglycosides (bactericidal) and linezolid (variable).

30S inhibitors

Aminoglycosides

“Buy at 30, ccel (sell) at 50.”

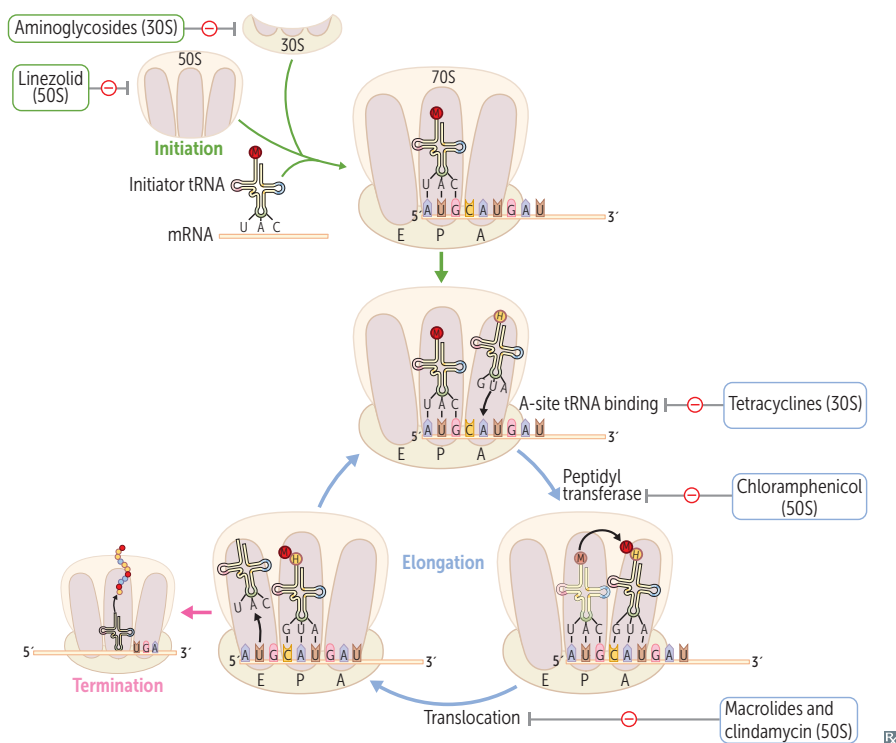
Tetracyclines

50S inhibitors

Chloramphenicol, Clindamycin

Erythromycin (macrolides)

Linezolid

**Aminoglycosides**

Gentamicin, Neomycin, Amikacin, Tobramycin, Streptomycin.

“Mean” (aminoglycoside) GNATS cannot kill anaerobes.

MECHANISM

Bactericidal; irreversible inhibition of initiation complex through binding of the 30S subunit. Can cause misreading of mRNA. Also block translocation. Require O₂ for uptake; therefore ineffective against anaerobes.

CLINICAL USE

Severe gram $\ominus$ rod infections. Synergistic with β -lactam antibiotics. Neomycin for bowel surgery.

ADVERSE EFFECTS

Nephrotoxicity, neuromuscular blockade (absolute contraindication with myasthenia gravis), ototoxicity (especially with loop diuretics), teratogenicity.

MECHANISM OF RESISTANCE

Bacterial transferase enzymes inactivate the drug by acetylation, phosphorylation, or adenylation.

Tetracyclines

Tetracycline, doxycycline, minocycline.

MECHANISM	Bacteriostatic; bind to 30S and prevent attachment of aminoacyl-tRNA. Limited CNS penetration. Doxycycline is fecally eliminated and can be used in patients with renal failure. Do not take tetracyclines with milk (Ca^{2+}), antacids (eg, Ca^{2+} or Mg^{2+}), or iron-containing preparations because divalent cations inhibit drugs' absorption in the gut.
CLINICAL USE	<i>Borrelia burgdorferi</i> , <i>M pneumoniae</i> . Drugs' ability to accumulate intracellularly makes them very effective against <i>Rickettsia</i> and <i>Chlamydia</i> . Also used to treat acne. Doxycycline effective against community-acquired MRSA.
ADVERSE EFFECTS	GI distress, discoloration of teeth and inhibition of bone growth in children, photosensitivity. "Teratocyclines" are teratogenic ; generally avoided in pregnancy and in children (except doxycycline).
MECHANISM OF RESISTANCE	↓ uptake or ↑ efflux out of bacterial cells by plasmid-encoded transport pumps.

Tigecycline

MECHANISM	Tetracycline derivative. Binds to 30S, inhibiting protein synthesis. Generally bacteriostatic.
CLINICAL USE	Broad-spectrum anaerobic, gram $\ominus$, and gram $\oplus$ coverage. Multidrug-resistant organisms (eg, MRSA, VRE).
ADVERSE EFFECTS	Nausea, vomiting.

Chloramphenicol

MECHANISM	Blocks peptidyltransferase at 50S ribosomal subunit. Bacteriostatic.
CLINICAL USE	Meningitis (<i>Haemophilus influenzae</i> , <i>Neisseria meningitidis</i> , <i>Streptococcus pneumoniae</i>) and rickettsial diseases (eg, Rocky Mountain spotted fever [<i>Rickettsia rickettsii</i>]). Limited use due to toxicity but often still used in developing countries because of low cost.
ADVERSE EFFECTS	Anemia (dose dependent), aplastic anemia (dose independent), gray baby syndrome (in premature infants because they lack liver UDP-glucuronosyltransferase).
MECHANISM OF RESISTANCE	Plasmid-encoded acetyltransferase inactivates the drug.

Clindamycin

MECHANISM	Blocks peptide transfer (translocation) at 50S ribosomal subunit. Bacteriostatic.
CLINICAL USE	Anaerobic infections (eg, <i>Bacteroides</i> spp., <i>C perfringens</i>) in aspiration pneumonia, lung abscesses, and oral infections. Also effective against invasive group A streptococcal infection. Treats anaerobic infections above the diaphragm vs metronidazole (anaerobic infections below diaphragm).
ADVERSE EFFECTS	Pseudomembranous colitis (<i>C difficile</i> overgrowth), fever, diarrhea.

Linezolid

MECHANISM	Inhibits protein synthesis by binding to the 23S rRNA of the 50S ribosomal subunit and preventing formation of the initiation complex.
CLINICAL USE	Gram $\oplus$ species including MRSA and VRE.
ADVERSE EFFECTS	Myelosuppression (especially thrombocytopenia), peripheral neuropathy, serotonin syndrome (due to partial MAO inhibition).
MECHANISM OF RESISTANCE	Point mutation of ribosomal RNA.

Macrolides

Azithromycin, clarithromycin, erythromycin.

MECHANISM	Inhibit protein synthesis by blocking trans location (“ macroslides ”); bind to the 50S ribosomal subunit. Bacteriostatic.
CLINICAL USE	Atypical pneumonias (<i>Mycoplasma</i> , <i>Chlamydia</i> , <i>Legionella</i>), STIs (<i>Chlamydia</i>), gram $\oplus$ cocci (streptococcal infections in patients allergic to penicillin), and <i>B pertussis</i> .
ADVERSE EFFECTS	MACRO : Gastrointestinal M otility issues, A rrhythmia caused by prolonged QT interval, acute C holestatic hepatitis, R ash, eO sinophilia. Increases serum concentration of theophylline, oral anticoagulants. Clarithromycin and erythromycin inhibit cytochrome P-450.
MECHANISM OF RESISTANCE	Methylation of 23S rRNA-binding site prevents binding of drug.

Polymyxins

Colistin (polymyxin E), polymyxin B.

MECHANISM	Cation polypeptides that bind to phospholipids on cell membrane of gram $\ominus$ bacteria. Disrupt cell membrane integrity $\rightarrow$ leakage of cellular components $\rightarrow$ cell death.
CLINICAL USE	Salvage therapy for multidrug-resistant gram $\ominus$ bacteria (eg, <i>P aeruginosa</i> , <i>E coli</i> , <i>K pneumoniae</i>). Polymyxin B is a component of a triple antibiotic ointment used for superficial skin infections.
ADVERSE EFFECTS	Nephrotoxicity, neurotoxicity (eg, slurred speech, weakness, paresthesias), respiratory failure.

Sulfonamides

Sulfamethoxazole (SMX), sulfisoxazole, sulfadiazine.

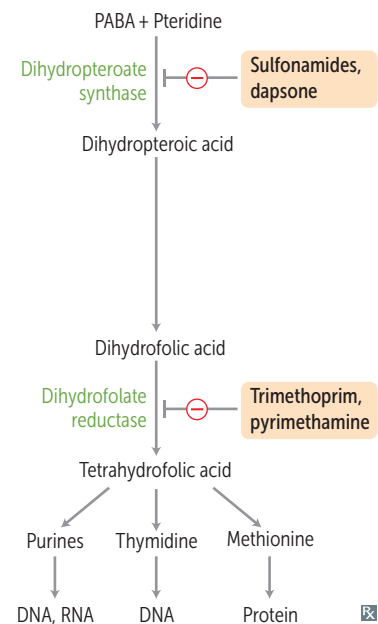
MECHANISM	Inhibit dihydropteroate synthase, thus inhibiting folate synthesis. Bacteriostatic (bactericidal when combined with trimethoprim).
CLINICAL USE	Gram $\oplus$, gram $\ominus$, <i>Nocardia</i> . TMP-SMX for simple UTI.
ADVERSE EFFECTS	Stevens-Johnson syndrome, urticaria, liver damage, folate deficiency, optic neuritis, nephrotoxicity, agranulocytosis, hemolysis if G6PD deficient, kernicterus in infants.
MECHANISM OF RESISTANCE	Altered enzyme (bacterial dihydropteroate synthase), $\downarrow$ uptake, or $\uparrow$ PABA synthesis.

Dapsone

MECHANISM	Similar to sulfonamides, but structurally distinct agent.
CLINICAL USE	Leprosy (lepromatous and tuberculoid), <i>Pneumocystis jirovecii</i> prophylaxis, or treatment when used in combination with TMP.
ADVERSE EFFECTS	Hemolysis if G6PD deficient, methemoglobinemia, agranulocytosis.

Trimethoprim

MECHANISM	Inhibits bacterial dihydrofolate reductase. Bacteriostatic.
CLINICAL USE	Used in combination with sulfonamides (trimethoprim-sulfamethoxazole [TMP-SMX]), causing sequential block of folate synthesis. Combination used for UTIs, <i>Shigella</i> , <i>Salmonella</i> , <i>Pneumocystis jirovecii</i> pneumonia treatment and prophylaxis, toxoplasmosis prophylaxis.
ADVERSE EFFECTS	Hyperkalemia (at high doses; similar mechanism as potassium-sparing diuretics), megaloblastic anemia, leukopenia, granulocytopenia, which may be avoided with coadministration of leucovorin (folinic acid). TMP Treats Marrow Poorly.



Fluoroquinolones

Ciprofloxacin, ofloxacin; respiratory fluoroquinolones: levofloxacin, moxifloxacin.

MECHANISM	Inhibit prokaryotic enzymes topoisomerase II (DNA gyrase) and topoisomerase IV. Bactericidal. Concurrent ingestion of divalent cations (eg, dairy, antacids) markedly decreases oral absorption.	
CLINICAL USE	Gram $\ominus$ rods of urinary and GI tracts (including <i>Pseudomonas</i>), some gram $\oplus$ organisms, otitis externa, community-acquired pneumonia.	
ADVERSE EFFECTS	GI upset, superinfections, skin rashes, headache, dizziness. Less commonly, can cause leg cramps and myalgias. Contraindicated during pregnancy or breastfeeding and in children < 18 years old due to possible damage to cartilage. Some may prolong QT interval.	May cause tendonitis or tendon rupture in people > 60 years old and in patients taking prednisone. Ciprofloxacin inhibits cytochrome P-450. Fluoroquinolones hurt attachments to your bones.
MECHANISM OF RESISTANCE	Chromosome-encoded mutation in DNA gyrase, plasmid-mediated resistance, efflux pumps.	

Daptomycin

MECHANISM	Lipopeptide that disrupts cell membranes of gram $\oplus$ cocci by creating transmembrane channels.	
CLINICAL USE	<i>S aureus</i> skin infections (especially MRSA), bacteremia, infective endocarditis, VRE.	Not used for pneumonia (avidly binds to and is inactivated by surfactant). “Dapto- myo-skin ” is used for skin infections but can cause myopathy .
ADVERSE EFFECTS	Myopathy, rhabdomyolysis.	

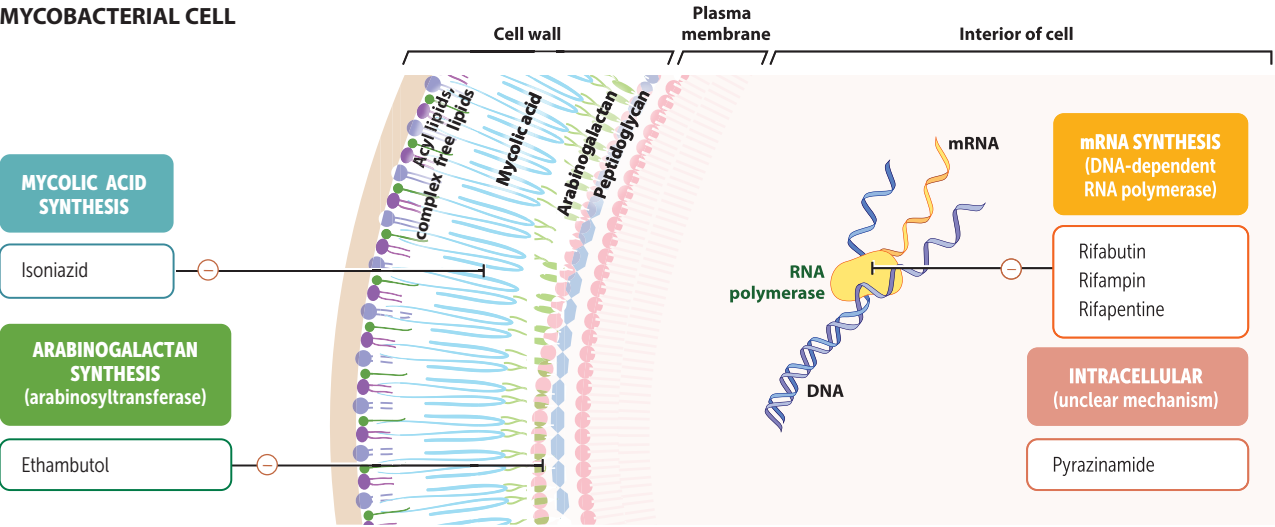
Metronidazole

MECHANISM	Forms toxic free radical metabolites in the bacterial cell that damage DNA. Bactericidal, antiprotozoal.	
CLINICAL USE	Treats G iardia, E ntamoeba, T richomonas, G ardnerella vaginalis, A naerobes (<i>Bacteroides</i> , <i>C difficile</i>). Can be used in place of amoxicillin in <i>H pylori</i> “triple therapy” in case of penicillin allergy.	GET GAP on the Metro with metronidazole ! Treats anaerobic infection below the diaphragm vs clindamycin (anaerobic infections above diaphragm).
ADVERSE EFFECTS	Disulfiram-like reaction (severe flushing, tachycardia, hypotension) with alcohol; headache, metallic taste.	

Antituberculous drugs

DRUG	MECHANISM	ADVERSE EFFECTS	NOTES
Rifamycins Rifampin, rifabutin, rifapentine	Inhibit DNA-dependent RNA polymerase → ↓ mRNA synthesis Rifamycin resistance arises due to mutations in gene encoding RNA polymerase	Minor hepatotoxicity, drug interactions (CYP450 induction), red-orange discoloration of body fluids (nonhazardous adverse effect)	Rifabutin favored over rifampin in patients with HIV infection due to less CYP450 induction Monotherapy rapidly leads to resistance
Isoniazid	Inhibits mycolic acid synthesis → ↓ cell wall synthesis Bacterial catalase-peroxidase (encoded by <i>katG</i>) is needed to convert INH to active form INH resistance arises due to mutations in <i>katG</i>	Vitamin B ₆ deficiency (peripheral neuropathy, sideroblastic anemia), hepatotoxicity, drug interactions (CYP450 inhibition), drug-induced lupus INH overdose can lead to seizures (often refractory to benzodiazepines)	Administer with pyridoxine (vitamin B ₆) INH Injures N eurons and H epatocytes (↑ risk of hepatotoxicity with ↑ age and alcohol overuse) Different INH half-lives in fast vs slow acetylators
Pyrazinamide	Mechanism uncertain	Hepatotoxicity, hyperuricemia	Works best at acidic pH (eg, in host phagolysosomes)
Ethambutol	Inhibits arabinosyltransferase → ↓ arabinogalactan synthesis → ↓ cell wall synthesis	Optic neuropathy (red-green color blindness or ↓ visual acuity, typically reversible)	Pronounce “ ey ethambutol”

MYCOBACTERIAL CELL



Antimycobacterial therapy

BACTERIUM	PROPHYLAXIS	TREATMENT
<i>M tuberculosis</i>	Rifamycin-based regimen for 3–4 months	Rifampin, Isoniazid, Pyrazinamide, Ethambutol (RIPE for treatment)
<i>M avium–intracellulare</i>	Azithromycin, rifabutin	Azithromycin or clarithromycin + ethambutol Can add rifabutin or ciprofloxacin
<i>M leprae</i>	N/A	Long-term treatment with dapsone and rifampin for tuberculoid form Add clofazimine for lepromatous form

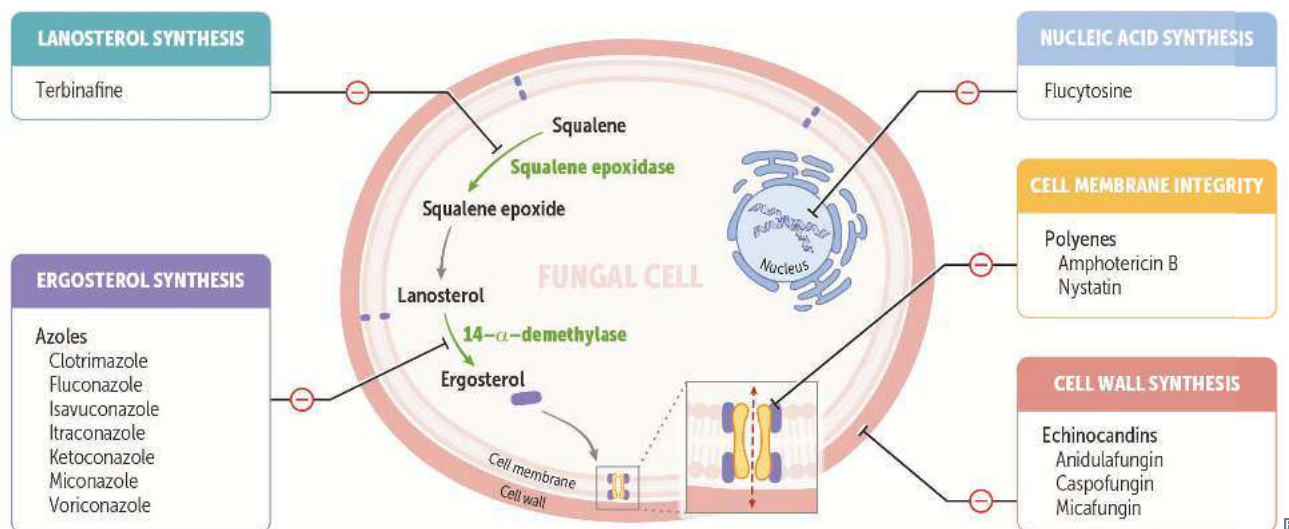
Antimicrobial prophylaxis

CLINICAL SCENARIO	MEDICATION
Exposure to meningococcal infection	Ceftriaxone, ciprofloxacin, or rifampin
High risk for infective endocarditis and undergoing surgical or dental procedures	Amoxicillin
History of recurrent UTIs	TMP-SMX
Malaria prophylaxis for travelers	Atovaquone-proguanil, mefloquine, doxycycline, primaquine, or chloroquine (for areas with sensitive species)
Pregnant patients carrying group B strep	Intrapartum penicillin G or ampicillin
Prevention of gonococcal conjunctivitis in newborn	Erythromycin ointment on eyes
Prevention of postsurgical infection due to <i>S aureus</i>	Cefazolin; vancomycin if ⊕ for MRSA
Prophylaxis of strep pharyngitis in child with prior rheumatic fever	Benzathine penicillin G or oral penicillin V

Prophylaxis in HIV infection/AIDS

CELL COUNT	PROPHYLAXIS	INFECTION
CD4+ < 200 cells/mm ³	TMP-SMX	<i>Pneumocystis</i> pneumonia
CD4+ < 100 cells/mm ³	TMP-SMX	<i>Pneumocystis</i> pneumonia and toxoplasmosis

Antifungal therapy



Amphotericin B

MECHANISM	Binds ergosterol (unique to fungi); forms membrane pores that allow leakage of electrolytes.	Amphotericin “ tears ” holes in the fungal membrane by forming pores.
CLINICAL USE	Serious, systemic mycoses. <i>Cryptococcus</i> (amphotericin B +/- flucytosine for cryptococcal meningitis), <i>Blastomyces</i> , <i>Coccidioides</i> , <i>Histoplasma</i> , <i>Candida</i> , <i>Mucor</i> . Intrathecally for coccidioidal meningitis.	Supplement K^+ and Mg^{2+} because of altered renal tubule permeability.
ADVERSE EFFECTS	Fever/chills (“shake and bake”), hypotension, nephrotoxicity, arrhythmias, anemia, IV phlebitis (“ amphoterrible ”).	Hydration ↓ nephrotoxicity. Liposomal amphotericin ↓ toxicity.

Nystatin

MECHANISM	Same as amphotericin B. Topical use only as too toxic for systemic use.
CLINICAL USE	“Swish and swallow” for oral candidiasis (thrush); topical for diaper rash or vaginal candidiasis.

Flucytosine

MECHANISM	Inhibits DNA and RNA biosynthesis by conversion to 5-fluorouracil by cytosine deaminase.
CLINICAL USE	Systemic fungal infections (especially meningitis caused by <i>Cryptococcus</i>) in combination with amphotericin B.
ADVERSE EFFECTS	Myelosuppression.

Azoles	Clotrimazole, fluconazole, isavuconazole, itraconazole, ketoconazole, miconazole, voriconazole.
MECHANISM	Inhibit fungal sterol (ergosterol) synthesis by inhibiting the cytochrome P-450 enzyme that converts lanosterol to ergosterol.
CLINICAL USE	Local and less serious systemic mycoses. Fluconazole for chronic suppression of cryptococcal meningitis in people living with HIV and candidal infections of all types. Itraconazole may be used for <i>Blastomyces</i> , <i>Coccidioides</i> , <i>Histoplasma</i> , <i>Sporothrix schenckii</i> . Clotrimazole and miconazole for topical fungal infections. Voriconazole for <i>Aspergillus</i> and some <i>Candida</i> . Isavuconazole for serious <i>Aspergillus</i> and <i>Mucor</i> infections.
ADVERSE EFFECTS	Testosterone synthesis inhibition (gynecomastia, especially with ketoconazole), liver dysfunction (inhibits cytochrome P-450), QT interval prolongation.

Terbinafine

MECHANISM	Inhibits the fungal enzyme squalene epoxidase.
CLINICAL USE	Dermatophytoses (especially onychomycosis—fungal infection of finger or toe nails).
ADVERSE EFFECTS	GI upset, headaches, hepatotoxicity, taste disturbance.

Echinocandins

	Anidulafungin, caspofungin, micafungin.
MECHANISM	Inhibit cell wall synthesis by inhibiting synthesis of β -glucan.
CLINICAL USE	Invasive aspergillosis, <i>Candida</i> .
ADVERSE EFFECTS	GI upset, flushing (by histamine release).

Griseofulvin

MECHANISM	Interferes with microtubule function; disrupts mitosis. Deposits in keratin-containing tissues (eg, nails).
CLINICAL USE	Oral treatment of superficial infections; inhibits growth of dermatophytes (tinea, ringworm).
ADVERSE EFFECTS	Teratogenic, carcinogenic, confusion, headaches, disulfiram-like reaction, $\uparrow$ cytochrome P-450 and warfarin metabolism.

Antiprotozoal therapy

	Pyrimethamine-sulfadiazine (toxoplasmosis), suramin and melarsoprol (<i>Trypanosoma brucei</i>), nifurtimox (<i>T cruzi</i>), sodium stibogluconate (leishmaniasis).
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Anti-mite/louse therapy

	Permethrin, malathion (acetylcholinesterase inhibitor), topical or oral ivermectin. Used to treat scabies (<i>Sarcoptes scabiei</i>) and lice (<i>Pediculus</i> and <i>Phthirus</i>).
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Chloroquine

MECHANISM	Blocks detoxification of heme into hemozoin. Heme accumulates and is toxic to plasmodia.
CLINICAL USE	Treatment of plasmodial species other than <i>P falciparum</i> (due to drug resistance from membrane pump that $\downarrow$ intracellular concentration of drug).
ADVERSE EFFECTS	Retinopathy (dependent on cumulative dose); pruritus (especially in dark-skinned individuals).

Anthelmintic therapy

Pyrantel pamoate, ivermectin, **mebendazole** (treats “**bendy** worms” by disrupting microtubules and cellular motility), praziquantel ($\uparrow$ Ca^{2+} permeability, $\uparrow$ vacuolization), diethylcarbamazine.

Oseltamivir, zanamivir

MECHANISM	Inhibit influenza neuraminidase $\rightarrow$ $\downarrow$ release of progeny virus.
CLINICAL USE	Treatment and prevention of influenza A and B. Beginning therapy within 48 hours of symptom onset may shorten duration of illness.

Baloxavir

MECHANISM	Inhibits the “cap snatching” (transfer of the 5′ cap from cell mRNA onto viral mRNA) endonuclease activity of the influenza virus RNA polymerase $\rightarrow$ $\downarrow$ viral replication.
CLINICAL USE	Treatment within 48 hours of symptom onset shortens duration of illness.

Remdesivir

MECHANISM	Prodrug of an ATP analog. The active metabolite inhibits viral RNA-dependent RNA polymerase and evades proofreading by viral exoribonuclease (ExoN) $\rightarrow$ $\downarrow$ viral RNA production.
CLINICAL USE	Recently approved for treatment of COVID-19 requiring hospitalization.

Acyclovir, famciclovir, valacyclovir

MECHANISM	Guanosine analogs. Monophosphorylated by HSV/VZV thymidine kinase and not phosphorylated in uninfected cells $\rightarrow$ few adverse effects. Triphosphate formed by cellular enzymes. Preferentially inhibit viral DNA polymerase by chain termination.
CLINICAL USE	No activity against CMV because CMV lacks the thymidine kinase necessary to activate guanosine analogs. Used for HSV-induced mucocutaneous and genital lesions as well as for encephalitis. Prophylaxis in patients who are immunocompromised. Also used as prophylaxis for immunocompetent patients with severe or recurrent infection. No effect on latent forms of HSV and VZV. Valacyclovir, a prodrug of acyclovir, has better oral bioavailability. For herpes zoster, use famciclovir.
ADVERSE EFFECTS	Obstructive crystalline nephropathy and acute kidney injury if not adequately hydrated.
MECHANISM OF RESISTANCE	Mutated viral thymidine kinase.

Ganciclovir

MECHANISM	Guanosine analog. 5′-monophosphate formed by a CMV viral kinase. Triphosphate formed by cellular kinases. Preferentially inhibits viral DNA polymerase.
CLINICAL USE	CMV, especially in patients who are immunocompromised. Valganciclovir, a prodrug of ganciclovir, has better oral bioavailability.
ADVERSE EFFECTS	Myelosuppression (leukopenia, neutropenia, thrombocytopenia), renal toxicity. More toxic to host enzymes than acyclovir.
MECHANISM OF RESISTANCE	Mutated viral kinase.

Foscarnet

MECHANISM	Viral DNA/RNA polymerase inhibitor and HIV reverse transcriptase inhibitor. Binds to pyrophosphate-binding site of enzyme. Does not require any kinase activation. Foscarnet = pyro fos phate analog.
CLINICAL USE	CMV retinitis in immunocompromised patients when ganciclovir fails; acyclovir-resistant HSV.
ADVERSE EFFECTS	Nephrotoxicity, multiple electrolyte abnormalities can lead to seizures.
MECHANISM OF RESISTANCE	Mutated DNA polymerase.

Cidofovir

MECHANISM	Preferentially inhibits viral DNA polymerase. Does not require phosphorylation by viral kinase.
CLINICAL USE	CMV retinitis in immunocompromised patients. Long half-life.
ADVERSE EFFECTS	Nephrotoxicity (coadminister cidofovir with probenecid and IV saline to ↓ toxicity).
MECHANISM OF RESISTANCE	Mutations in the viral DNA polymerase gene.

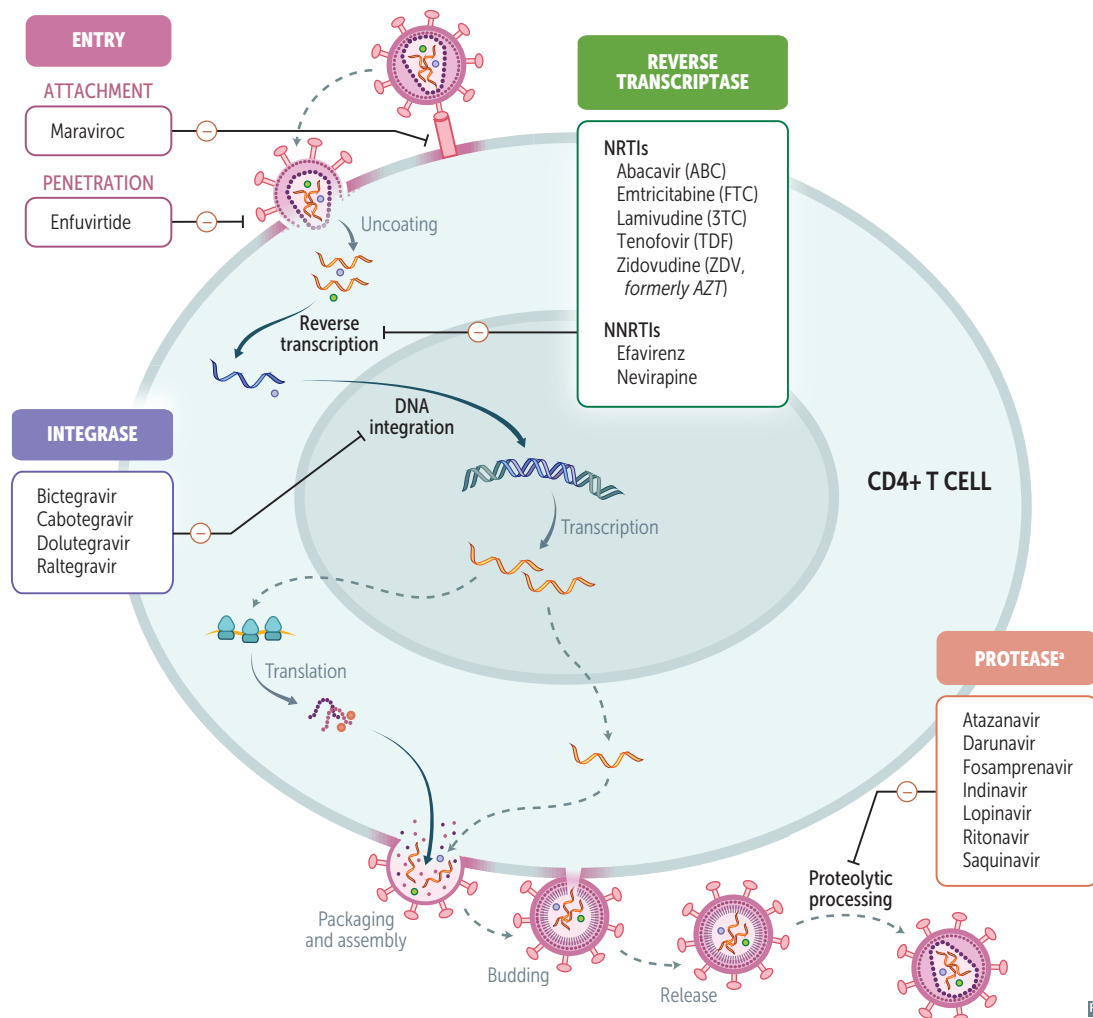
HIV therapy

Antiretroviral therapy (ART): often initiated at the time of HIV diagnosis.
 Strongest indication for use with patients presenting with AIDS-defining illness, low CD4+ cell counts (< 500 cells/mm³), or high viral load. Regimen consists of 3 drugs to prevent resistance: 2 NRTIs and preferably an integrase inhibitor.
 Most ARTs are active against both HIV-1 and HIV-2 (exceptions: NNRTIs and enfuvirtide not effective against HIV-2).
 Tenofovir + emtricitabine can be administered as pre-exposure prophylaxis.

DRUG	MECHANISM	ADVERSE EFFECTS
NRTIs		
Abacavir (ABC) Emtricitabine (FTC) Lamivudine (3TC) Tenofovir (TDF) Zidovudine (ZDV, formerly AZT)	Competitively inhibit nucleotide binding to reverse transcriptase and terminate the DNA chain (lack a 3' OH group). Tenofovir is a nucle T ide; the others are nucleosides. All need to be phosphorylated to be active. ZDV can be used for general prophylaxis and during pregnancy to ↓ risk of fetal transmission. Have you dined (vudine) with my nuclear (nucleosides) family?	Myelosuppression (can be reversed with granulocyte colony-stimulating factor [G-CSF] and erythropoietin), nephrotoxicity. Abacavir contraindicated if patient has HLA-B*5701 mutation due to ↑ risk of hypersensitivity.
NNRTIs		
Doravirine Efavirenz Rilpivirine	Bind to reverse transcriptase at site different from NRTIs. Do not require phosphorylation to be active or compete with nucleotides.	Rash and hepatotoxicity are common to all NNRTIs. Vivid dreams and CNS symptoms are common with efavirenz.
Integrase strand transfer inhibitors		
Bictegravir Dolutegravir	Also called integrase inhibitors. Inhibit HIV genome integration into host cell chromosome by reversibly inhibiting HIV integrase.	↑ creatine kinase, weight gain.

HIV therapy (continued)

DRUG	MECHANISM	ADVERSE EFFECTS
Protease inhibitors		
Atazanavir Darunavir Lopinavir Ritonavir	Prevents maturation of new virions. Maturation depends on HIV-1 protease (<i>pol</i> gene), which cleaves the polypeptide products of HIV mRNA into their functional parts. All protease inhibitors require boosting with either ritonavir or cobicistat. Navir (never) tease a protease .	Hyperglycemia, GI intolerance (nausea, diarrhea). Rifampin (potent CYP/UGT inducer) ↓ protease inhibitor concentrations; use rifabutin instead. Ritonavir (cytochrome P-450 inhibitor) is only used as a boosting agent.
Entry inhibitors		
Enfuvirtide	Binds gp41, inhibiting viral entry. Enfuvirtide inhibits fusion .	Skin reaction at injection sites.
Maraviroc	Binds CCR-5 on surface of T cells/monocytes, inhibiting interaction with gp120. Maraviroc inhibits docking .	



^aAll protease inhibitors require boosting with either ritonavir (protease inhibitor only used as a boosting agent) or cobicistat (cytochrome P450 inhibitor).

Hepatitis C therapy

Chronic HCV infection treated with multidrug therapy that targets specific steps within HCV replication cycle (HCV-encoded nonstructural proteins). Examples of drugs are provided.

DRUG	MECHANISM	TOXICITY
NS5A inhibitors		
Elbasvir	Inhibits NS5A, a viral phosphoprotein that plays a key role in RNA replication	Headache, diarrhea
Ledipasvir		
Pibrentasvir	Exact mechanism unknown	
Velpatasvir		
NS5B inhibitors		
Sofosbuvir	Inhibits NS5B, an RNA-dependent RNA polymerase acting as a chain terminator Prevents viral RNA replication	Fatigue, headache
NS3/4A inhibitors		
Glecaprevir	Inhibits NS3/4A, a viral protease, preventing viral replication	Headache, fatigue
Grazoprevir		
Alternative drugs		
Ribavirin	Inhibits synthesis of guanine nucleotides by competitively inhibiting IMP dehydrogenase	Hemolytic anemia, severe teratogen

Disinfection and sterilization

Goals include the reduction of pathogenic organism counts to safe levels (disinfection) and the inactivation of all microbes including spores (sterilization).

Autoclave ^a	Pressurized steam at > 120°C. May not reliably inactivate prions.
Alcohols	Denature proteins and disrupt cell membranes.
Chlorhexidine	Disrupts cell membranes and coagulates intracellular components.
Chlorine ^a	Oxidizes and denatures proteins.
Ethylene oxide ^a	Alkylating agent.
Hydrogen peroxide ^a	Free radical oxidation.
Iodine and iodophors	Halogenation of DNA, RNA, and proteins. May be sporicidal.
Quaternary amines	Impair permeability of cell membranes.

^aSporicidal.

Antimicrobials to avoid in pregnancy

ANTIMICROBIAL	ADVERSE EFFECTS
Sulfonamides	Kernicterus
Aminoglycosides	Ototoxicity
Fluoroquinolones	Cartilage damage
Clarithromycin	Embryotoxic
Tetracyclines	Discolored teeth, inhibition of bone growth
Ribavirin	Teratogenic
Griseofulvin	Teratogenic
Chloramphenicol	Gray baby syndrome
Safe children take really good care.	